

A wallet is not a person.

One human, one credit line. World ID proof of personhood reaches Creditcoin through the Attestcoin Protocol, a zero-knowledge proof is verified on Creditcoin itself, and a verified human gets an uncollateralized credit line that follows the person, not the wallet.

- Creditcoin exists to give credit history to people the banking system cannot see.
- Every uncollateralized design on the chain scores a wallet, and a wallet is free.
- Ten wallets, ten perfect histories, then a default on the eleventh at full size.



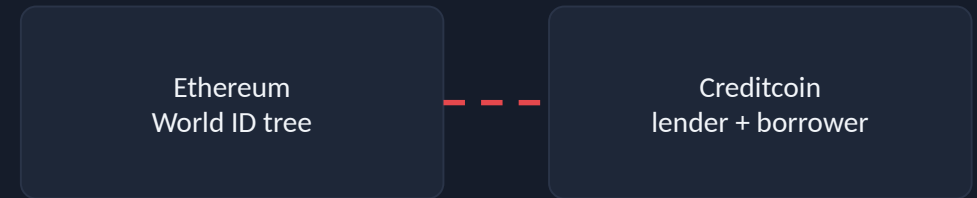
Uncollateralized credit needs a unique borrower

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The market is real. The missing piece is not the ledger.

- Aella records loans on Creditcoin through Credal and serves more than a million users.
- A lender without collateral has one question: will this person come back, and is there one of them.
- On-chain history answers the first. Nothing on Creditcoin answers the second.
- World ID has Orb-verified humans in Kenya, Argentina, Indonesia, the Philippines, Brazil and Malaysia. Those are Creditcoin's markets.
- But World ID's identity tree lives on Ethereum, and Creditcoin cannot see it.

The two ways across today



~~trusted bridge~~

~~trusted oracle~~

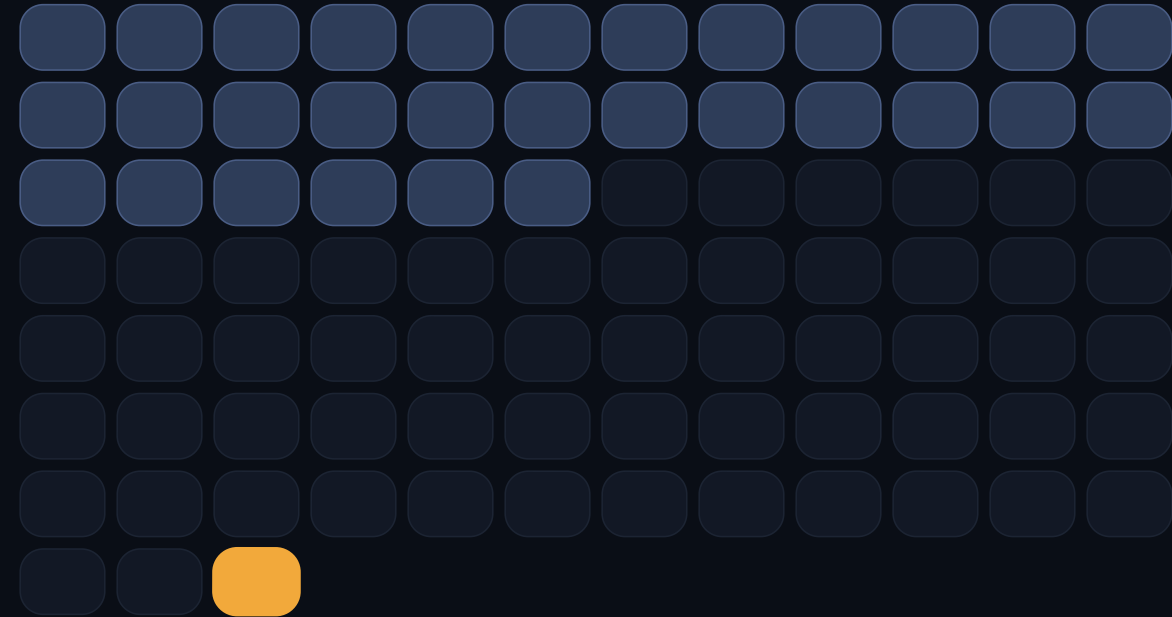
Both are the exact thing the Attestcoin Protocol was built to remove. So we removed them.

The 30-passport trap

87 submissions in this hackathon. About 30 build the same thing.

- Read loan repayments on Sepolia, mint a score, call it a credit passport.
- Ten more are AI guardrails. Nine are escrow on proof.
- Zero touch identity. Zero use zero-knowledge proofs. Zero do proof of personhood.
- Every one of those passports can be forged by generating a new wallet. Not hacked. Generated.
- A perfect score that means nothing, because the thing being scored is not the thing that owes the money.

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30 credit passports

Humanline

0 identity · 0 zero-knowledge · 0 proof of personhood

The nullifier is the human

Bind the person. Never bind the key.

- A World ID nullifier comes from an Orb iris scan and an action string. It is the same value whatever wallet the person uses.
- Every piece of state in Humanline is keyed by it. Nothing is keyed by an address.
- Re-bind to a new wallet and the identity, the limit and the history all move with the person.
- Default, and the freeze attaches to the nullifier. It survives a new wallet, a new key, a new device.
- The forgery that breaks every other credit passport costs nothing. Forging this costs a second iris.

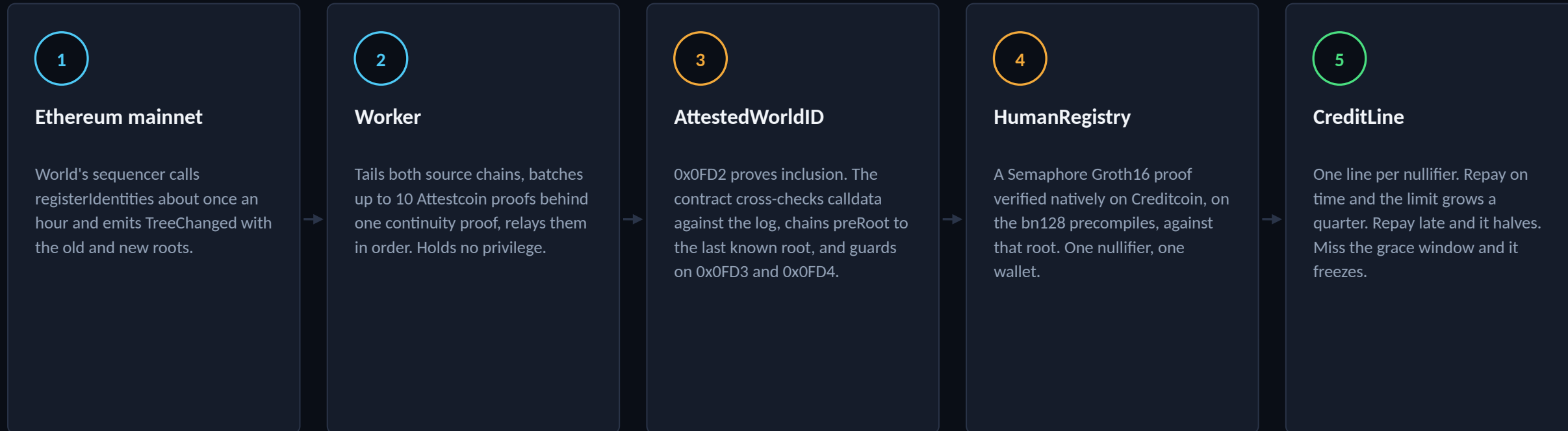


the freeze is on the hash, not on any address

How it works

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Roots come in through Attestcoin. Proofs are checked on Creditcoin. Credit follows the human.



No owner. No pause. No upgrade path. Anyone can run the relay, and the contracts do not care who does.

Attestcoin depth: ten surfaces, all load-bearing

Surface	Where it runs	Why it is load-bearing
verifyAndEmit (0x0FD2)	ASCBBase.execute	No root reaches the verifier without it
Batch verification	getBatchProof, executeBatch	Up to 10 proofs under one continuity proof
Calldata decoding	decodeCommonTxFields(tx).data	Yields postRoot and the count of humans added
Log decoding	getLogsByEventSignature	The root itself
Emitter and status binding	to, log.address_, status	Rejects look-alike events and reverted txs
calculateTxIndex	queryId, sourceTxIndex	Replay key and ordering evidence
ChainInfo (0x0FD3)	Finality depth guard	Accepted only 32 blocks behind the attested tip
AttestorStash (0x0FD4)	Quorum floor	Refuses roots attested by a thin set
Two source chains	chainKey 3 and chainKey 1	Production and judge-reproducible paths
Zero-knowledge on top	Semaphore over a relayed root	Without Attestcoin the verifier has no root

Remove Attestcoin and Humanline has no roots, so no humans, so no credit. There is no degraded mode.

What is real

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Say the boring part out loud. It is what makes the rest credible.

Real, on chain, right now

- Ethereum mainnet World ID roots, produced by World's own sequencer about once an hour, relayed unattended.
- An Attestcoin proof verified by the 0x0FD2 precompile inside every relay transaction. No mock, no canned fixture.
- Groth16 Semaphore verification executing on Creditcoin CC3, on the bn128 precompiles, in one block.
- Sepolia staging roots and World simulator proofs, so a judge reproduces the whole flow without an Orb.

Testnet only, and not claimed

- hUSD is a test stablecoin we mint. Lender deposits are testnet funds.
- The demo deployment uses minute-long loan terms so a full cycle fits in a video.
- Attestcoin cannot prove a payment did not happen. We never pretend otherwise.
- A default is declared from a passed deadline plus the absence of a repayment on Creditcoin, which is native state, not a cross-chain absence claim.

Every green claim above links to a transaction on Etherscan or Blockscout from the live app.

Demo

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Three minutes, six steps, every step an on-chain artifact.

1 /relay

0:28

A real Ethereum root landing on Creditcoin. Click through to both transactions and read the humans-added count decoded from the calldata.

2 /app

1:05

Connect a Creditcoin wallet, verify with the World simulator, watch registration confirm in one block. The wallet becomes a human.

3 Credit

1:45

Open a line at 25 hUSD, borrow 20, repay, limit grows to 31.25. Every action links to Blockscout.

4 Re-bind

2:25

A second wallet, the same identity. The line and history follow the person. A second line is refused with LineExists.

5 /judge

2:40

Run the negative-path suite live. Every attack rejected by a named error.

6 Counter

2:52

N real roots relayed from Ethereum mainnet, M humans in the registry, zero trusted parties.

Live at <https://humanline.credit>

· demo video linked from the submission

What Attestcoin proves, and what we add

Readability proofs are a primitive. A correct application still supplies its own ordering, binding and replay key.

Attestcoin proves

inclusion + continuity

That is all it proves, and it is enough.

Everything else is our job. Ten checks, in this order:

1. receipt status

2. source contract

3. emitter binding

4. calldata vs log

5. root chaining

6. finality depth

7. attester quorum

8. replay by queryId

9. root expiry

10. nullifier uniqueness

Attacks, and the error that stops them

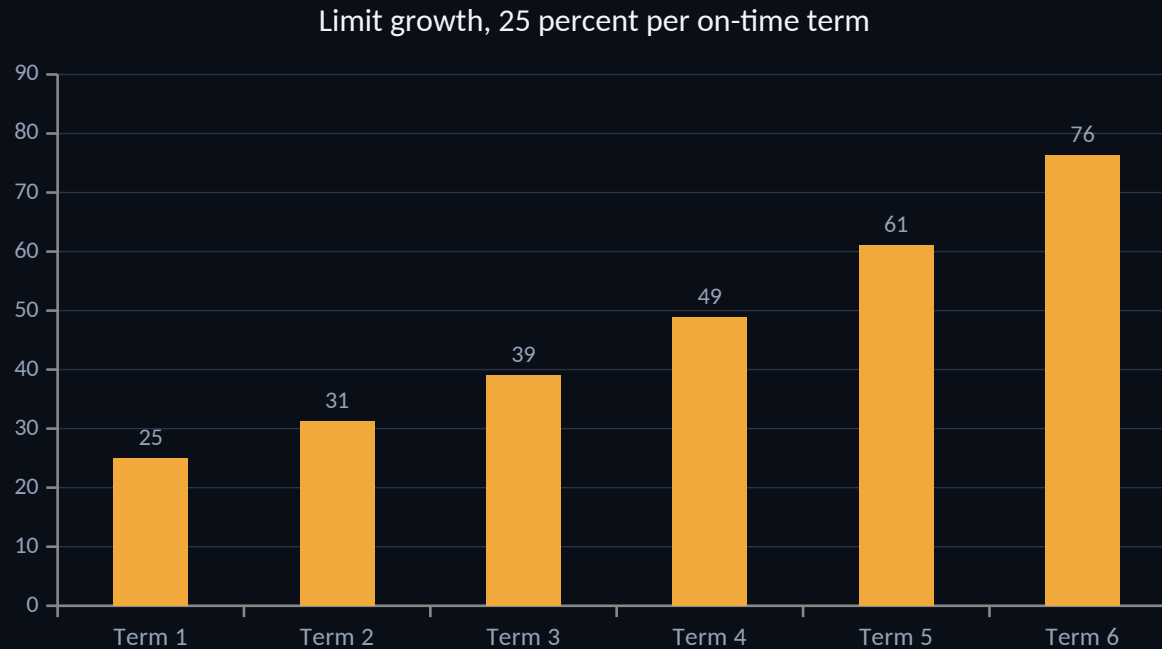
Forged root or proof	<code>rejected at 0xFD2, first</code>
Decoy <code>TreeChanged</code> from another contract	<code>skipped by the emitter filter</code>
Two genuine <code>TreeChanged</code> logs in one tx	<code>AmbiguousTreeChange</code>
Calldata that disagrees with the log	<code>CalldataLogMismatch</code>
A genuine root, out of sequence	<code>UnknownPreRoot</code>
A root shallow enough to be reorged	<code>NotFinal, 32 blocks of depth</code>
A thin or captured attester set	<code>ThinQuorum, floor of 3</code>
Replaying a proof already relayed	<code>queryId already processed</code>
Stealing a proof from the mempool	<code>ProofInvalid, signal is caller</code>
A defaulted human with a fresh wallet	<code>LineFrozen, on the nullifier</code>

No admin keys. Every constant is an immutable. Anyone can relay.

Market and business

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A personhood layer with a lending business on top, not a lending app with an identity feature.



One borrower, first year, modeled from the deployed contract parameters. Not measured.

281 USD

cumulative draw across six terms

3.76 USD

revenue from one first-year borrower

1 percent

origination, plus a spread on the term fee

- Three lines: a 0.10 USD verification fee from the lender, 1 percent origination, and a spread on the pool.
- The compounding asset is the shared default record. Ten lenders on one registry is a credit bureau nobody operates.
- Every integration, every repayment and every root is Attestcoin volume.

Roadmap, and Periscope

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Every step adds Attestcoin volume instead of replacing it.

1

Month 1

AttestedWorldID and HumanRegistry on Creditcoin mainnet, published as a public read interface. First lender using isHuman as a sybil check through Credal. Two independent relayers.

2

Month 3

Lines funded by PenguinSwap LPs. Repay-from-Ethereum: a USDC transfer proven through Attestcoin and credited on Creditcoin. BSC as a source chain when chainKey 8 ships.

3

Month 6

Supervised pilot in Kenya and Argentina with a lending partner, and a published cohort loss curve for personhood-gated credit. Nobody has one.

Periscope

World ID 4.0 verification lives on World Chain, an OP-Stack rollup whose output roots are posted to Ethereum. Attestcoin already attests Ethereum. Prove the posting and you have proven the rollup.

This is not a Humanline feature. It gives Attestcoin reach into every OP-Stack rollup through a chain it already covers. We are just the first application that needs it.

The ask

12

The top three teams enter the CEIP fast-track. We are asking for that, and for three things through it.

1

Capital

A 6-month supervised pilot in Kenya and Argentina. The deliverable is a published cohort loss curve every Creditcoin lender can underwrite against.

2

A lender introduction

To Aella, or any Credal lender. The registry is one read call from being useful to a loan book that already exists.

3

Engineering support

Build Periscope with the Creditcoin team, bringing World Chain and every other OP-Stack rollup under Attestcoin.

15

real World ID roots relayed

600

World ID identities carried

1,000

hUSD in the lender pools

0

trusted parties

One human, one credit line.

<https://humanline.credit>
<https://github.com/rajkaria/humanline>